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Culbertson et al.

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(54) **RADIATION SHIELD WITH MAGNETIC PROPERTIES**

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(58) **Field of Classification Search**

USPC 250/515.1, 516.1, 517.1, 519.1
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,864,124	A *	2/1975	Breton		B01J 37/00	156/89.28
4,766,608	A	8/1988	Cusick			
5,525,408	A	6/1996	Weir			
5,745,925	A	5/1998	Ghilardi			
5,908,884	A	6/1999	Kawamura			
6,869,683	B2 *	3/2005	Sakurai		B32B 25/20	428/447
7,274,031	B2	9/2007	Smith			
7,815,820	B2 *	10/2010	Tan		H05K 9/0083	250/515.1
2010/0176317	A1 *	7/2010	Smith		B32B 1/08	250/518.1

* cited by examiner

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(57) **ABSTRACT**

A radiation attenuation shield, method, and system are disclosed. The shield includes a polymer, a radiation attenuating material, and a magnetic material. The radiation attenuating material and the magnetic material may be dispersed within the polymer to form an attenuation layer. Further, a magnetic material layer may be positioned adjacent to the attenuation layer or encase the attenuation layer. The radiation attenuation shield may be made by combining the components to create a mixture and then inserting the mixture in a mold until a solidified shape is formed. Moreover, the radiation attenuation shield of the present invention may be mechanically secured to a structure to contain radiation. Further, the shield may be secured to a structure by using the magnetic properties of the shield.

33 Claims, 5 Drawing Sheets